

# Optimizing Areal Density of Heat-Assisted Interlaced Magnetic Recording

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Heat-assisted interlaced magnetic recording (HIMR) increases areal density by alternating narrow top tracks with wide bottom tracks. Most of the gain comes from higher linear density of the bottom tracks, which require higher laser current. The laser current elevates head temperature and degrades reliability, limiting HIMR adoption. This work develops a spin-stand method to optimize HIMR areal density as a function of laser and write current. Results show that the areal density gain does not always require the highest laser current. The maximal areal density is achieved using a 15% increase in laser current and a 25% increase in write current. Reducing the laser current increase from 15% to 5% lowers the areal density gain only slightly, from 9.5% to 8.6%. This low sensitivity is attributed to saturation of spatial signal-to-noise ratio of the bottom track confined between two top tracks and enables a favorable balance between density gain and reliability.

*Index Terms*— Heat-assisted magnetic recording, Interlaced magnetic recording.

## I. INTRODUCTION

HEAT-ASSISTED magnetic recording (HAMR) technology has been leading the areal density capability (ADC) of hard disk drives. HAMR supports three types of track layouts: conventional magnetic recording (CMR), shingled magnetic recording (SMR), and interlaced magnetic recording (IMR) [1]. In CMR, each track can be accessed randomly. In SMR, a band of tracks are written sequentially with overlay between neighboring tracks; this layout increases the track density at the expense of write accessibility because the entire block of tracks need to be re-written to update a single data sector. In IMR, the tracks alternate between wide bottom tracks and narrow top tracks; this layout increases the linear density over CMR with a smaller penalty in write accessibility than SMR: one data sector in the bottom track can be updated by re-writing the bottom track and the two adjacent top tracks.

Compared to CMR and SMR, heat-assisted IMR (HIMR) has a disadvantage in reliability because the bottom tracks require higher laser current. Higher laser current increases the temperature of the heads which degrades the lifetime exponentially. This reliability concern hinders the adoption of HIMR. One potential approach to reduce the laser current is to use higher writer current to record the bottom track.

In this study, a spin stand test has been developed to map the HIMR ADC gain as a function of writer current and laser current. The results show that ADC gain saturates at higher laser current, and higher write current enables similar ADC gain at lower laser current. This trade-off allows HIMR to realize significant ADC gain with less than 10% increase in laser current. We have also assessed the signal power and spatial noise of the bottom track to explain the saturation of the ADC gain.

## II. EXPERIMENTAL DETAILS

The method to maximize the ADC of CMR has been described in Ref. [2]. The optimal laser current and write current (60 mA for the head design used in this study) for CMR

are used to write the HIMR top tracks. The track pitch of HIMR tracks is kept the same as that of CMR tracks.

The HIMR bottom tracks are written with two different write current (60 and 75 mA), and five different laser current levels (optimal laser current for CMR multiplied by 1, 1.05, 1.1, 1.15, and 1.2).

At each combination of write current and laser current, the linear density of the bottom track is varied until the sector failure rate (SFR) of the bottom track sandwiched between two top tracks reaches 15%. The linear density of the top track is then varied until the bit error rate (BER) of the top track sandwiched between two bottom tracks reaches  $\sim 2.0$  decade. The ADC for HIMR is then calculated by the average critical linear densities of the bottom and top tracks multiplied by the track density, which is the same with CMR.

To assess limiting factors for the linear density gain of the bottom tracks, the spin stand test based on synchronous write and pseudo-random bit sequence (PRBS) waveform analysis [2] has been modified to assess the signal power and spatial noise power of the bottom tracks as a function of write current and laser current.

## III. RESULTS

ADC results of one head are summarized in Fig. 1. Fig. 1a shows the linear density capability  $L_b$  of the bottom track as a function of effective laser current  $I_b$ . The denominators  $L_0$  and  $I_0$  are the linear density and effective laser current at the maximal CMR ADC. The circles and diamonds are measured with write current of 60 and 75 mA, respectively. When using the same laser current and write current as CMR, the linear density capability of the bottom track is 106.8% that of the CMR track. The increase is from the reduction of the number of adjacent track writes from 6 for CMR to 1 for HIMR. As the normalized laser current increases from 100% to 120%, the normalized linear density increases from 106.8% to 113.4%. When using 75 mA writer current for the bottom track, the normalized linear density is 109.3% at the same laser current, and increases to 114.7% at high laser current.

Figure 1b shows the linear density capability of the top track as a function of laser current and write current of the bottom tracks. When using the same write current and laser current as CMR to write the bottom track, the linear density of the top track is 105.8% that of the CMR track. The increase is from the relaxing of the BER criteria from -2.25 decade for CMR to -2.0 decade for HIMR. As the laser current of the bottom track increases, the normalized linear density of the top track drops slightly.

The normalized linear densities of the bottom and top tracks are averaged to calculate the normalized ADC of HIMR, and the results of this head are shown in Fig. 1c. When using writer current of 60 mA for the bottom track, the normalized ADC increases from 106.3% to 109.2% with laser current. With writer current of 75 mA, the normalized ADC increases from 107.5% to 109.5%. Using higher writer current increases the HIMR ADC for the same bottom track laser current.

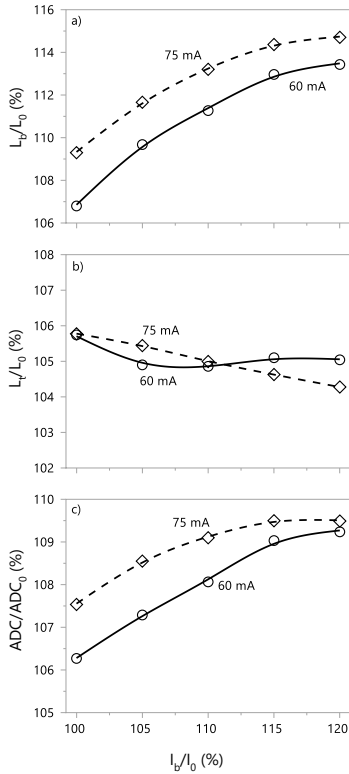


Fig. 1. Linear densities of the bottom track (a) and top track (b), and ADC (c) as a function of laser current and write current used to write the bottom tracks.  $L_b$  and  $L_t$  are the linear density of the bottom and top tracks, respectively.  $L_0$  is the linear density capability for CMR.  $I_b$  is the laser current used to write the bottom track.  $I_0$  is the optimal laser current for CMR.  $ADC_0$  is the CMR ADC. Two levels of write current, 60 mA and 75 mA, are used to write the bottom tracks. The write current for the top tracks is 60 mA.

The maximal HIMR ADC for this head is achieved at writer current of 75 mA, and normalized laser current of 115%, i.e., 15% higher laser current than CMR. Increasing the normalized laser current to 20% results in similar ADC. This saturation in ADC at 15% higher laser current occurs for multiple heads. The saturation of ADC with laser current is caused by the combination of nonlinear increase of linear density of the bottom track, and the slight decrease of the linear density of the top track.

The decrease of the linear density of the top track with laser current used to write the bottom tracks is caused by the increased side reading from the stronger bottom tracks.

To assess the nonlinear increase of the bottom track linear density with laser current, the waveforms of signal and noise of the bottom track have been measured using the synchronous write adjacent track test. The results show that the signal power of the bottom track sandwiched between two top tracks increases slightly with laser current, and the spatial noise decreases with laser current. Both the signal and spatial noise saturate at 110% laser current and only slightly improve after that. This saturation is because that bottom tracks are confined by the top tracks, and the signal-noise-ratio increase only comes from the increase of the quality of the long marks and transitions instead of the increase of the track width.

#### IV. CONCLUSION

We have shown that HIMR ADC gain is nonlinear with bottom track laser current increase. The root cause is a combination of the increased side reading from the better-defined bottom tracks, and the saturation of the quality of the bottom track confined between two top tracks with laser current. As a result, reducing the laser current increase from 20% to 5% only decreases the ADC gain from 9.5% to 8.6%, but will significantly improve the head reliability.

#### REFERENCES

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